

The equations of motion may be expressed in terms of the Lagrangian as:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\mathbf{q}}} = \frac{\partial L}{\partial \mathbf{q}}.$$

(Note that this must be interpreted in general as a set of $N_c = 3N$ equations, one for each atomic coordinate.) The real usefulness of the Lagrangian perspective comes when we introduce different types of coordinates to express the equations of motion; examples of this are given below.

When we introduce a smooth change of variables $\mathbf{q} = \boldsymbol{\Phi}(\mathbf{Q})$, where $\boldsymbol{\Phi} : \mathbb{R}^{N_c} \rightarrow \mathbb{R}^{N_c}$, this induces a corresponding transformation of the velocity vector by

$$\dot{\mathbf{q}} = \boldsymbol{\Phi}'(\mathbf{Q})\dot{\mathbf{Q}},$$

where $\boldsymbol{\Phi}'$ is the $N_c \times N_c$ Jacobian matrix of $\boldsymbol{\Phi}$. Then the Lagrangian of the system is transformed to

$$\tilde{L} = \frac{\dot{\mathbf{Q}}^T \boldsymbol{\Phi}'(\mathbf{Q})^T \mathbf{M} \boldsymbol{\Phi}'(\mathbf{Q}) \dot{\mathbf{Q}}}{2} - U(\boldsymbol{\Phi}(\mathbf{Q})),$$

It is not essential for the change of variables to be a map from $\mathbb{R}^{N_c}$ to $\mathbb{R}^{N_c}$. For example, in a system with constraints, the change of variables might represent a parameterization of the manifold defined by the constraints. Then the mapping $\boldsymbol{\Phi}$ might be from $\mathbb{R}^{N_d}$ to $\mathbb{R}^{N_c}$, where $N_c \geq N_d$. An abstract set of configuration variables is referred to as a set of *generalized coordinates*. The introduction of generalized coordinates in an N-body system typically leads to a Lagrangian which can be viewed as quadratic in the velocities but with a “configuration-dependent mass matrix.” We will assume that any such changes of variables are regular transformations in the sense that $\boldsymbol{\Phi}'$ is of full rank and the resulting generalized mass matrix is invertible.

Although generalized coordinates are often useful for understanding molecular systems, as we shall learn in later chapters, it is desirable to avoid the use of formulations with a configuration-dependent mass matrix since it complicates the design of numerical methods with good conservation properties. This is one reason that molecular simulation methods are typically described in a Cartesian coordinate framework.

1.4 Hamiltonian Formulation

In this section, we introduce the Hamiltonian description of the motion which is useful for discussing the underlying geometric structure associated to classical mechanics. We can understand the Hamiltonian formulation as a natural description of classical mechanics from several perspectives. First, it is based directly on the

most basic conservation law associated to a closed system (the conservation of energy) as it is formulated in terms of the energy function. Second, it is the starting point for the quantum mechanical treatment of a system of particles, since the Schrödinger equation is written in terms of the quantum analogue of the energy function. Third, we may view the Hamiltonian as the *Legendre transformation* of the Lagrangian viewed as a function of the velocities. This is a *conjugacy* relationship which ensures that the properties of the two systems are, under typical conditions, equivalent.

Abstractly, a Legendre transformation of a given convex function $g = g(\xi) : \mathbb{R}^m \rightarrow \mathbb{R}$ is a new function $\tilde{g} = \tilde{g}(\eta) : \mathbb{R}^m \rightarrow \mathbb{R}$ defined by

$$\tilde{g}(\eta) = \sup_{\xi} (\eta^T \xi - g(\xi)),$$

where the symbol $\sup_{\xi}$ means to find the supremum of the given expression with respect to variation of the vector ξ . In the case of the Lagrangian $L(\mathbf{q}, \mathbf{v}) = \mathbf{v}^T \mathbf{M}(\mathbf{q}) \mathbf{v} / 2 - U(\mathbf{q})$ (which we have seen is the formulation of a mechanical system in generalized coordinates) we find

$$\sup_{\mathbf{v}} (\mathbf{p}^T \mathbf{v} - (\mathbf{v}^T \mathbf{M}(\mathbf{q}) \mathbf{v} / 2 - U(\mathbf{q}))) = \sup_{\mathbf{v}} (\mathbf{p}^T \mathbf{v} - \mathbf{v}^T \mathbf{M}(\mathbf{q}) \mathbf{v} / 2) + U(\mathbf{q}).$$

If $\mathbf{M}(\mathbf{q})$ is invertible, the supremum is achieved precisely when

$$\mathbf{v} = \mathbf{M}^{-1}(\mathbf{q}) \mathbf{p},$$

which gives the standard definition of the momentum vector $\mathbf{p}$ in terms of velocities,

$$\mathbf{p} = \frac{\partial L}{\partial \dot{\mathbf{q}}} = \mathbf{M}(\mathbf{q}) \dot{\mathbf{q}},$$

and, with $\mathbf{v} = \mathbf{M}^{-1}(\mathbf{q}) \mathbf{p}$, the Legendre transformation results in a new function

$$H(\mathbf{q}, \mathbf{p}) \stackrel{\text{def}}{=} \mathbf{p}^T \mathbf{M}(\mathbf{q})^{-1} \mathbf{p} / 2 + U(\mathbf{q}).$$

This is precisely the energy function, written in terms of positions and momenta. In this form it is termed the *Hamiltonian* and it provides an alternative description of the system. The equations of motion can be written

$$\dot{\mathbf{q}} = \frac{\partial H}{\partial \mathbf{p}}, \quad \dot{\mathbf{p}} = -\frac{\partial H}{\partial \mathbf{q}}.$$

For constant $\mathbf{M}$ we obtain the dynamical equations,

$$\dot{\mathbf{q}} = \mathbf{M}^{-1} \mathbf{p}, \quad \dot{\mathbf{p}} = \mathbf{F} = -\partial U / \partial \mathbf{q}.$$

This is not a big step from the Newtonian form of the equations of motion $\dot{\mathbf{q}} = \mathbf{v}; \mathbf{M}\dot{\mathbf{v}} = \mathbf{F}$. More generally, for molecular models, the Hamiltonian and Lagrangian formulations are interchangeable, but the use of the Hamiltonian form is preferred for allowing simplified description of the geometric character of the solutions of the system as we discuss in Chaps. 2–4.

The set of all positions and momenta for which the energy is finite is termed the *phase space*. The instantaneous state of a molecular system involving many, say N , particles moving in $\mathbb{R}^3$ is described by coordinates and positions, i.e., by a point in $\mathbb{R}^{6N}$. The “points” defining atomic positions (generally taken to be located in $\mathbb{R}^3$) should be clearly distinguished from the “points,” in $\mathbb{R}^{6N}$, of the phase space, which represent the collection of all position and momenta vectors of the atoms. One may speak of a “phase space point” or simply “phase point” to describe the latter. We are ultimately going to be interested in how sets of phase points are carried forward by solution of the differential equations. It is crucial to observe that when we discuss such sets the individual points therein are understood to be completely independent of one another, and their motion is governed only by the laws of mechanics as described by the equations of motion.

1.5 Existence and Uniqueness of Solutions

One of the most important properties of a typical classical molecular Hamiltonian system is the existence and uniqueness of solutions started from a generic initial condition. The traditional study of existence and uniqueness for systems of nonlinear ordinary differential equations gives local results only, but for Hamiltonian systems considered in molecular dynamics the solutions are typically globally defined. The uniqueness of solutions is easily verified in the usual way (as for the local result for uniqueness of solutions). The key point is that, with the energy constraint, solutions typically remain bounded for all time.

Assume that U is a potential energy function which is bounded below, $U \geq U_{\min}$. For given $E_0 \geq U_{\min}$ define $\Sigma_{E_0} = \{(\mathbf{q}, \mathbf{p}) | H(\mathbf{q}, \mathbf{p}) = E_0\}$. Then, for $(\mathbf{q}, \mathbf{p}) \in \Sigma_{E_0}$ we have

$$\frac{\mathbf{p}^T \mathbf{M}^{-1} \mathbf{p}}{2} + U(\mathbf{q}) = E_0 \Rightarrow \frac{\mathbf{p}^T \mathbf{M}^{-1} \mathbf{p}}{2} = E_0 - U(\mathbf{q}) \leq E_0 - U_{\min}.$$

$\mathbf{M}^{-1}$ is a positive definite matrix (assumed here to be constant), so we can infer that the momenta are bounded at fixed total energy. We would like to say something similar for positions. We have, at energy E_0 ,

$$U_{\min} \leq U(\mathbf{q}) \leq E_0.$$

What is needed is an assumption that the level sets $\hat{\Sigma}_\alpha = \{\mathbf{q} | U(\mathbf{q}) = \alpha\}$ are bounded uniformly for $\alpha \in [U_{\min}, E_0]$. Then it follows that solutions satisfying the energy constraint remain confined to a compact (closed and bounded) set.

The assumption on U is not satisfied by some simple potentials. For example consider $U(x, y) = x^2$ which is completely independent of y and so places no restriction on that variable for constant energy. The condition on U means that it has a *confining* property. To ensure bounded solutions in systems that do not have a confining potential, we should add other assumptions, for example the assumption that the position vectors of each atom are restricted by periodic boundary conditions to lie on a three-dimensional torus (see Section 1.6).

1.5.1 The Flow Map

Consider now the initial value problem

$$\dot{\mathbf{z}} = \mathbf{f}(\mathbf{z}), \quad \mathbf{z}(0) = \boldsymbol{\xi}, \quad (1.5)$$

in a m -dimensional space. If we assume that $\mathbf{f}$ corresponds to a molecular Hamiltonian system satisfying the assumptions of the existence and uniqueness result of the previous subsection, then we may define a mapping from a point in phase space to the point t units later along the time-evolution starting from the initial point. We refer to this map as the *flow map* and denote it by $\mathcal{F}_t$. $\mathcal{F}_t(\boldsymbol{\xi}) = \mathbf{z}(t)$ solves the initial value problem (1.5). It may equally describe the evolution of a point with time, or the evolution of a set of points with time.

The classical systems treated here can be solved forward or backward in time. Observe that $\mathcal{F}_{-t}\mathcal{F}_t = \text{Id}$ (the identity map), thus the flow map is invertible and, indeed, the family of flow maps defined for different values of t form an Abelian group under the operation of composition ($\mathcal{F}_t\mathcal{F}_s = \mathcal{F}_s\mathcal{F}_t = \mathcal{F}_{t+s}$). The flow map of a Hamiltonian system conserves its Hamiltonian, thus

$$H(\mathcal{F}_t(\boldsymbol{\xi})) \equiv H(\boldsymbol{\xi}).$$

(This equation just expresses the conservation of energy.)

The flow map provides an easy notation for describing solutions (even in cases where it is difficult or impossible to compute it exactly). When we speak of numerical methods for solving differential equations, we generally mean methods that approximate the flow map in a suitable sense.

Despite the existence of a well-defined flow map, we cannot solve most systems of differential equations analytically (even Hamiltonian ones) so we cannot usually write out a formula for the flow map. An exception is the harmonic oscillator,

$$\dot{q} = p, \quad \dot{p} = -\Omega^2 q,$$